

My capstone project will be a terminal user interface (TUI) audio player. The program will allow users to navigate and play local music files through an interactive layout with visual menus, rather than a standard command-line prompt. It is intended for users who want a minimalist way to listen to music in a terminal workflow without switching windows or using a heavy GUI application. This program is useful because it combines file management with real-time audio control in a highly visual way. The user will enter file paths to their music, and the program will extract metadata or allow for manual entry. Users will interact with the program by navigating a menu using arrow keys or mouse clicks to play, skip, or adjust the volume while the program tracks the playback state in a persistent header.

A major challenge will be concurrency, ensuring the audio plays in the background while the TUI continues to refresh the time elapsed bar and listen for new user input. I will also need to handle the logic of linking the song nodes to the actual file paths on the hard drive to ensure the correct file is loaded. Managing the TUI layout while manipulating playlist pointers will also be complex.

I will use an abstract base class, `UIElement`, to define how visual components are rendered. From this, I will use inheritance to define classes such as `SongList`, `ProgressBar`, and `Menu`. This allows me to use polymorphism via a virtual `handleInput` function, so the main loop can send keyboard or mouse events to any UI component and execute the correct logic based on the object type.

For the playlist, I will implement a doubly linked list from scratch. This is a more advanced data structure than standard arrays or vectors and is necessary to traverse songs both forward and backward using pointers.

Finally, I will use recursion for a directory crawler. When a user provides a music folder, a recursive function will search through all subfolders to find and link every supported audio file to the master library.